



MOCK EXAMINATION

 Module: **Database Administration (COMP 30021.1)**

 Level: **3** Max. Marks: **100** Duration: **1 ¼ Hours**

ID NUMBER									

Instructions to Student:

- Answer *any two* questions. Each question carries 50 marks.

The following LOs are achieved by the student by completing the assessment successfully.

Manage database instances and database storage

Work with Distributed Databases.

- You have been entrusted with the development of a comprehensive database system. Design the database schema and provide the SQL commands for the following tasks:
 - Create a tablespace specifically for storing the tables of this database. Ensure that the tablespace command includes the size of the datafile and automatic segment space management.
 - Define the tables as described below. Assume appropriate data types and constraints.

 Patients: Patient_ID (PK), First_Name, Last_Name, Date_of_Birth, Gender, Contact_Number
 Doctors: Doctor_ID (PK), First_Name, Last_Name, Specialization, Contact_Number, Email
 Appointments: App_ID (PK), Patient_ID (FK), Doctor_ID (FK), Appointment_Date, Status
 - Populate (Insert the values in table...Add records) all tables with at least 2 sample records each.
 - A company is using a **RAID 5 array with 3 disks** to store critical data with fault tolerance. In RAID 5, data is striped across two disks and **parity information** is stored on the third disk. Parity is calculated using the XOR operation: The following 6-bit data blocks are stored on each disk:
Disk 1: Data A = **101101**
Disk 2: Data B = **011110**
 Demonstrate **RAID 5 Reconstruction with this example.**

(50 Marks)

2.

a. A company wants to maximize the performance of its analytics processing system, which requires very high read and write speeds. To achieve this, they decide to configure a RAID 0 (striping) array. They have selected 5 disks, each with a capacity of 2 TB and a speed of 200 MB/s for both read.

1. Calculate the total usable storage capacity of the RAID 0 array.
2. If the company needs a minimum of 12 TB of usable storage, how many additional disks (of the same capacity) must they add to the RAID 0 array?

(25 Marks)

b. Explain Distributed Database Partitioning and any two types of the same.

(25 Marks)

3.

a) Discuss four characteristics of Distributed Databases

(25 Marks)

b) Explain advantages of **Distributed Databases**

(25 Marks)